

HW15

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Packages

We load our usual packages: `tidyverse` as a general tool, and `infer` to help with random sampling.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
#read in data
```

```
data <- read_csv("data/data.csv")
```

```
## Rows: 2044 Columns: 5
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (2): degree, grass
```

```
## dbl (3): hrsrelax, mntllhth, hrs1
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(data, 5)
```

```
## # A tibble: 5 x 5
```

```
##   hrsrelax mntllhth  hrs1 degree      grass
```

```
##   <dbl>     <dbl> <dbl> <chr>     <chr>
```

```
## 1      2       3    55 BACHELOR    <NA>
```

```
## 2      4       6    45 BACHELOR    LEGAL
```

```
## 3     NA      NA    NA LT HIGH SCHOOL <NA>
```

```
## 4     NA      NA    NA LT HIGH SCHOOL NOT LEGAL
```

```
## 5     NA      NA    NA LT HIGH SCHOOL NOT LEGAL
```

1.b) Interpretation Sentence: The dataset comes from a survey that collected information about individuals mental health, relaxation habits, work hours, education level, and whether marijuana is legal where they live. It was collected to better understand how lifestyle factors and social policies might be related to health and well-being. This study is observational because researchers recorded existing traits and behaviors without manipulating any variables. Because it is observational, we can find associations but cannot establish

causation. Additionally since the dataset is a sample of individuals, we can generalize results to the larger population only if the sample was randomly selected and is representative. Otherwise no we cannot generalize to a larger population.

1.c.pt1) I decided to choose the hrsrelax variable to focus on for the numerical variable.

```
# remove NA's before doing assignment
hrsrelax_clean <- na.omit(data$hrsrelax)
```

1.c.pt2)

```
# description of data / summary info
summary(hrsrelax_clean)
```

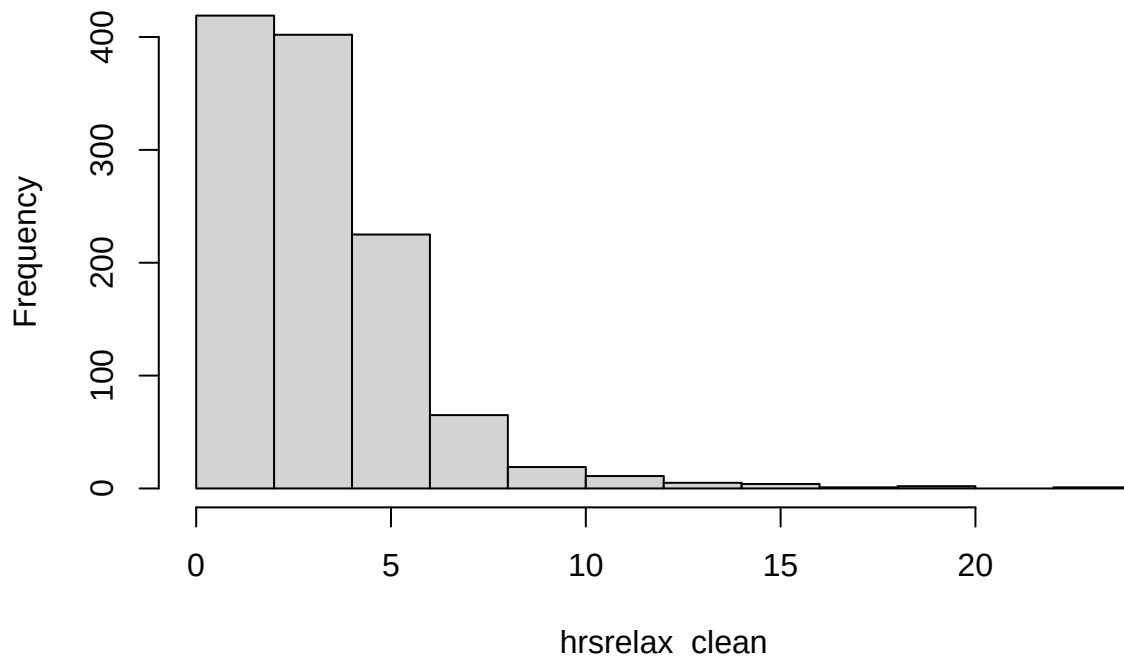
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00   2.00   3.00   3.68   5.00   24.00
```

```
sd(hrsrelax_clean)
```

```
## [1] 2.629641
```

```
hist(hrsrelax_clean)
```

Histogram of hrsrelax_clean



1.c.pt3) calculate a 95% and a 99% confidence interval for the mean. for my reference: Confidence Interval= $\bar{x} \pm z_{\text{star}} (\text{SE})$

note $\bar{x}$ = sample mean SE = $s/\sqrt{n}$ s = sample standard dev n = sample size z_{star} = crit val

defining vars

```
n <- length(hrsrelax_clean) #sample size
mean_hrsrelax <- mean(hrsrelax_clean) # sample mean
sd_hrsrelax <- sd(hrsrelax_clean) #sample SD
se_hrsrelax <- sd_hrsrelax / sqrt(n)
```

getting 95% and 99% confidence intervals

```
# 95% CI
z_star_95 <- 1.96
lower_95 <- mean_hrsrelax - z_star_95 * se_hrsrelax
upper_95 <- mean_hrsrelax + z_star_95 * se_hrsrelax
```

```
lower_95
```

```
## [1] 3.52852
```

```
upper_95
```

```
## [1] 3.831965
```

```
# -----
# 99% CI
z_star_99 <- 2.576
lower_99 <- mean_hrsrelax - z_star_99 * se_hrsrelax
upper_99 <- mean_hrsrelax + z_star_99 * se_hrsrelax
```

```
lower_99
```

```
## [1] 3.480836
```

```
upper_99
```

```
## [1] 3.879649
```

ANS below: result for 95% confidence interval(rounded 4 decimal places): (3.5285,3.8320)

result for 99% confidence interval(rounded 4 decimal places): (3.4808,3.8796)

Sentence of interpretation: Based on the sample we are 95% confident that the true mean number of hours relaxed is between 3.5285 and 3.8320 hours. We are 99% confident that the true mean number of hours relaxed is between 3.4808 and 3.8796 hours.

compute 85% confidence interval:

```
z_star_85 <- qnorm(1 - (1-0.85)/2) #gives you the z-critical value needed to build a confidence interval
lower_85 <- mean_hrsrelax - z_star_85 * se_hrsrelax
upper_85 <- mean_hrsrelax + z_star_85 * se_hrsrelax
```

```
lower_85
```

```
## [1] 3.568809
```

```
upper_85
```

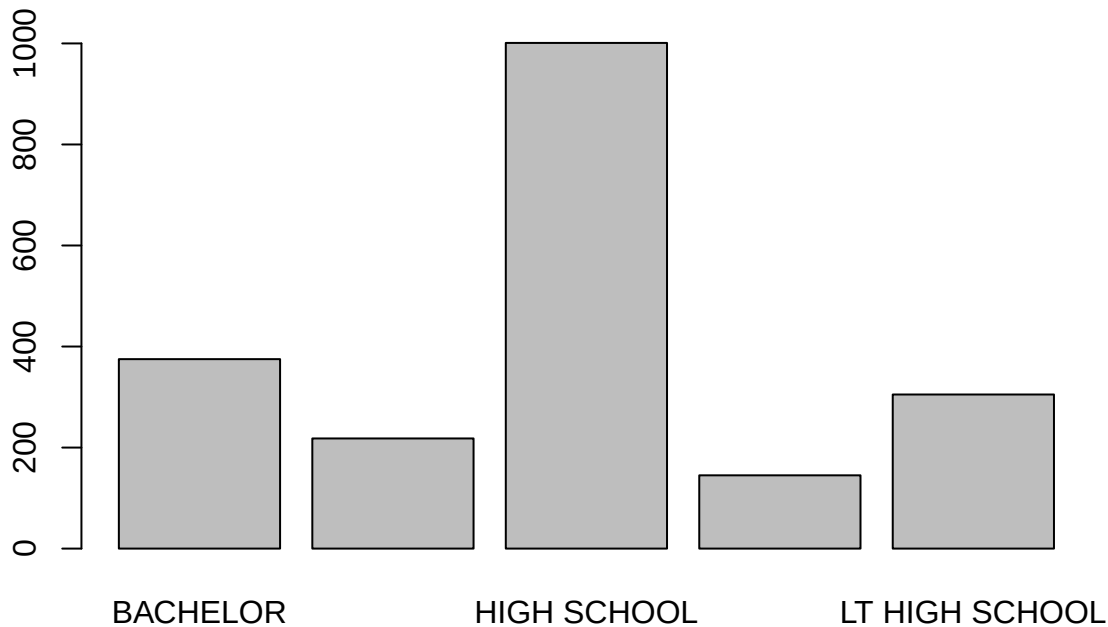
```
## [1] 3.791676
```

Sentence of interpretation: Based on the sample, we are 85% confident that the true mean number of hours people relax is between 3.5688 and 3.7917 hours.

#———— Categorical variable part

```
#choose categorical and drop all na vals
degree_clean <- na.omit(data$degree)
table(degree_clean) # shows the number of people with each education level
```

```
## degree_clean
##      BACHELOR      GRADUATE    HIGH SCHOOL JUNIOR COLLEGE LT HIGH SCHOOL
##           375           218           1001           145           305
barplot(table(degree_clean)) # not sure if this was required thought to include it anyway
```



note: I plan to focus on a specific proportion: (individuals with a Bachelors degree) getting that from chunk below

```
prop_bachelor <- mean(degree_clean == "BACHELOR")
prop_bachelor
```

```
## [1] 0.1834638
```

```
#get sample size + get standard error
n_degree <- length(degree_clean)
n_degree
```

```
## [1] 2044
```

```
se_prop_bachelor <- sqrt(prop_bachelor * (1 - prop_bachelor) / n_degree)
se_prop_bachelor
```

```
## [1] 0.00856096
```

```
#compute 95% confidence intervals
z_star_95 <- 1.96
```

```
lower_95_bachelor <- prop_bachelor - z_star_95 * se_prop_bachelor
upper_95_bachelor <- prop_bachelor + z_star_95 * se_prop_bachelor
```

```
lower_95_bachelor
```

```
## [1] 0.1666843
```

```
upper_95_bachelor
```

```
## [1] 0.2002433
```

Interpretation for 95% confidence interval

Based on the sample, we are 95% confident that the true proportion of people with a Bachelor's degree in the population is between 0.1666843 and 0.2002433

#compute 99% confidence intervals

```
z_star_99 <- 2.576

lower_99_bachelor <- prop_bachelor - z_star_99 * se_prop_bachelor
upper_99_bachelor <- prop_bachelor + z_star_99 * se_prop_bachelor

lower_99_bachelor

## [1] 0.1614108
upper_99_bachelor
```

```
## [1] 0.2055168
```

Interpretation for 99% confidence interval

Based on the sample, we are 99% confident that the true proportion of people with a Bachelor's degree in the population is between 0.1614108 and 0.2055168

#compute 85% confidence intervals

```
z_star_85 <- qnorm(1 - (1 - 0.85)/2)

lower_85_bachelor <- prop_bachelor - z_star_85 * se_prop_bachelor
upper_85_bachelor <- prop_bachelor + z_star_85 * se_prop_bachelor

lower_85_bachelor

## [1] 0.17114
upper_85_bachelor
```

```
## [1] 0.1957876
```

Interpretation for 85% confidence interval

Based on the sample, we are 85% confident that the true proportion of people with a Bachelor's degree in the population is between 0.17114 and 0.1957876.

question 2 section answers

2. Before a big election, polling companies regularly contact random samples of likely voters, trying to predict the proportion of people who will vote for each candidate. Imagine an election for governor of a state with two major candidates—call them “A” and “B”.
 - a. In a typical competitive election, a major candidate will have the support of about 50% of the voters. Use this expectation to estimate the margin of error (at 95% confidence) for estimating the proportion of support for candidate A using results a poll based on a random sample of 1000 likely voters. (The margin of error at 95% confidence is equal to twice the standard error. It's the special name often used in polling for the number that we add and subtract from our point estimate to make a 95% confidence interval.) Show your work.

```
# Margin of error formula for proportion at 95% confidence:
# ME = 2 * SE
p_hat <- 0.5
```

```
n_poll_1000 <- 1000

se_poll_1000 <- sqrt(p_hat * (1 - p_hat) / n_poll_1000)
margin_of_error_1000 <- 2 * se_poll_1000

margin_of_error_1000

## [1] 0.03162278
```

Based on a sample size of 1000, the margin of error is approximately 3.16%.

- b. Suppose that we wanted to get more precise polling results by using a larger sample in our poll. What would the estimated margin of error be, if we doubled the sample size to 2000 people? Again, show your work.

```
n_2000 <- 2000

se_2000 <- sqrt(p_hat * (1 - p_hat) / n_2000)
margin_error_2000 <- 2 * se_2000

margin_error_2000

## [1] 0.02236068
```

Based on a sample size of 2000, the margin of error is approximately 2.24%.

- c. How large a sample would you recommend to make the margin of sampling error (at 95% confidence) less than 1 percentage point? Again, show your work.

```
target_margin <- 0.01

required_n <- (p_hat * (1 - p_hat)) / ( (target_margin / 2)^2 )

required_n

## [1] 10000
```

To achieve a margin of error less than 1%, a sample size of approximately 10,000 voters would be needed.